Intro to OWL Reasoning with Protégé

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ISWC 2026

Who is Mark Wallace?

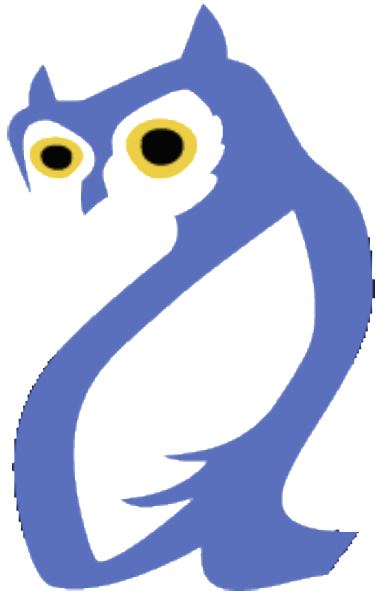
- Expert Ontologist & KG Engineer
- Decades of experience building KG systems
- Semantic Web since 2004
- Invited speaker in Semantic Web since 2009
- Consultant with Semantic Arts since 2017
- Built Large-scale RDF/OWL applications
  - Fortune 100, Start-ups, US Government



What We'll Learn

- Some **key elements** of **OWL** reasoning
- How to **code** them using Protégé
- How to **test** them using Protégé

- What I've found **most useful** in **real IT** systems



Why We'll Learn

- Reasoning is **cool**!

But also, in this resurgent AI era…

- LLMs do better with **knowledge graphs** for **context**
- *Semantic* Graphs (with OWL) make **meaning** more **explicit**
- OWL Reasoning can **simplify** AI's work
  - Because more **direct** connections in the graph
  - Less LLM "guessing", **hallucination**



OWL Axioms (and those we'll cover today)

Class Definition:
subClassOf
intersectionOf
unionOf
complimentOf
oneOf
disjointWith

Property Characteristics:
ObjectProperty
DatatypeProperty
subPropertyOf
domain & range
inverseOf
TransitiveProperty
SymmetricProperty
FunctionalProperty
InverseFunctionalProperty
Asymmetric
reflexive
Disjoint properties

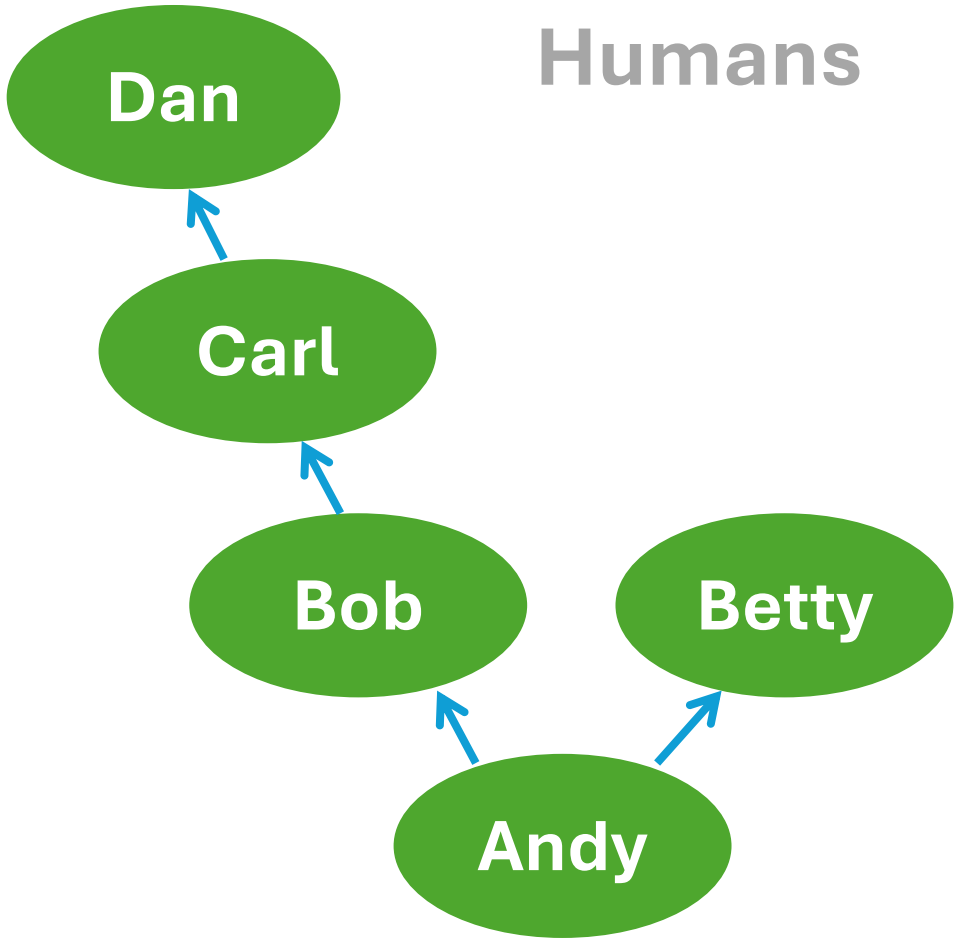
Class Property Restrictions:
allValuesFrom
someValuesFrom
hasValue

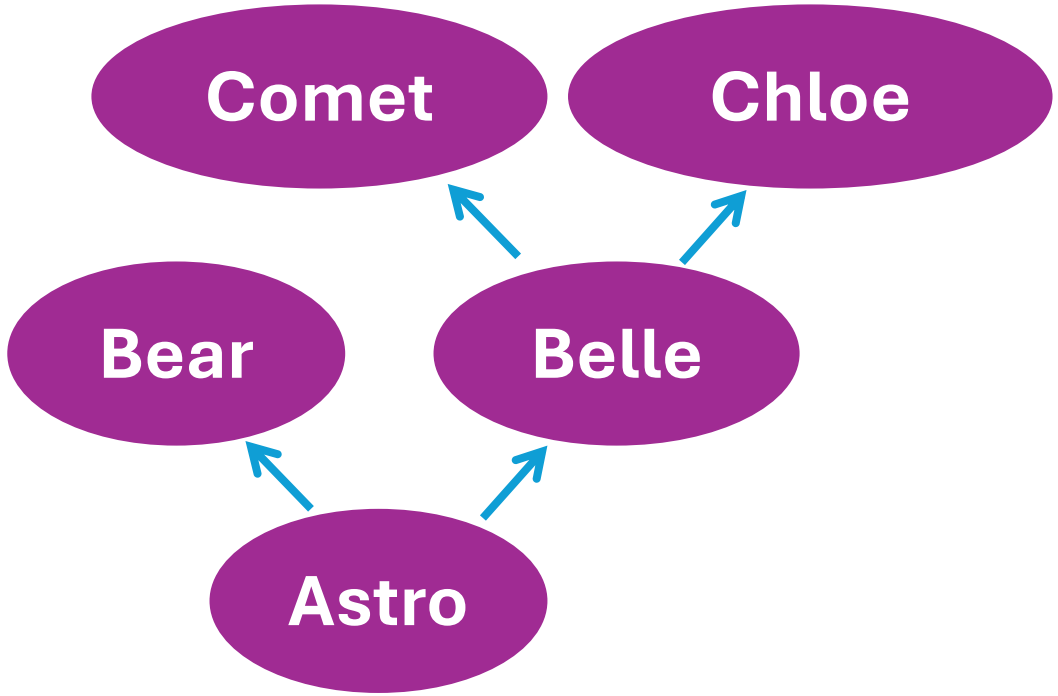
Restricted Cardinality:
minCardinality
maxCardinality
cardinality

Equality/Inequality:
equivalentClass
equivalentProperty
sameAs; ***differentFrom***
AllDifferent
distinctMembers

AND MORE!
keys
property chains
rich datatypes, data ranges

Today's Scenario: Ancestry





Do the work!

github.com/marksem/bari

What We Learned Today!

Class Definition:
subClassOf
intersectionOf
unionOf
disjointWith (for classes)

Property Characteristics:
ObjectProperty
subPropertyOf
domain & range
TransitiveProperty
FunctionalProperty

Class Property Restrictions:
someValuesFrom
hasValue

Equality/Inequality:
equivalentClass
differentFrom (for individuals)

Questions?



Thank You for Joining!

